

# Differential Equations: Exponential & Logistic Growth — Problem Set

## Warm-up

1. Verify that  $y = Ce^{kt}$  is a solution to the differential equation  $\frac{dy}{dt} = ky$ .
2. Verify that  $T = 20 + Ae^{-kt}$  is a solution to the differential equation  $\frac{dT}{dt} = -k(T - 20)$ .
3. A population  $P$  grows according to  $\frac{dP}{dt} = 0.05P$ . If the initial population is 1000, find  $P$  when  $t = 10$ .
4. An object cools according to  $\frac{dT}{dt} = -k(T - 25)$ . If  $T = 100$  when  $t = 0$ , find the constant  $A$  in the solution  $T = 25 + Ae^{-kt}$ .
5. If  $\frac{dN}{dt} = kN$ , and  $N$  doubles in 5 years, find the exact value of  $k$ .
6. A substance decays such that  $\frac{dM}{dt} = -0.1M$ . Find the half-life of the substance.
7. Show that  $P = \frac{100}{1+9e^{-t}}$  satisfies  $P = 10$  when  $t = 0$ .
8. For the logistic equation  $\frac{dP}{dt} = 0.2P(1 - \frac{P}{5000})$ , identify the carrying capacity.  
(Note: The Carrying Capacity is the value of  $P$  when  $\frac{dP}{dt} = 0$ )

## Skill-building

1. If  $T = S + (T_0 - S)e^{-kt}$ , find the limit of  $T$  as  $t \rightarrow \infty$ .
2. A container of liquid at  $80^\circ\text{C}$  is placed in a room with a constant temperature of  $20^\circ\text{C}$ . After 10 minutes, the liquid is  $60^\circ\text{C}$ . Use  $\frac{dT}{dt} = -k(T - 20)$  to find the temperature after 20 minutes.
3. The population of a colony of bacteria grows at a rate proportional to the population  $P$ . If the population increases from 500 to 1500 in 3 hours, find the population after 6 hours.
4. A particle moves such that its velocity  $v$  satisfies  $\frac{dv}{dt} = -2(v - 5)$ . If  $v = 20$  when  $t = 0$ , find  $v$  in terms of  $t$ .
5. Solve the differential equation  $\frac{dy}{dx} = 3y - 12$  given  $y(0) = 10$ .
6. A tank contains 1000 litres of brine. The amount of salt  $S$  changes according to  $\frac{dS}{dt} = -kS$ . If 10% of the salt is removed every 2 hours, find  $k$ .
7. The growth of a population  $N$  is modelled by  $\frac{dN}{dt} = k(1000 - N)$ . If  $N(0) = 200$  and  $N(5) = 500$ , find  $N(t)$ .
8. Show that  $y = \frac{L}{1+Be^{-kLt}}$  is a solution to the logistic equation  $\frac{dy}{dt} = ky(L - y)$ .
9. A rumor spreads through a school of 1000 students according to  $\frac{dP}{dt} = 0.001P(1000 - P)$ , where  $P$  is the number of students who have heard the rumor. If 10 students know the rumor at  $t = 0$ , find how many know it at  $t = 5$ .

10. A fish farm has a carrying capacity of 10,000 fish. The population  $F$  follows  $\frac{dF}{dt} = rF(1 - \frac{F}{10000})$ . If the population grows from 1000 to 2000 in 1 year, find  $r$ .
11. An object is heated to  $100^{\circ}\text{C}$  and then kept in a medium at  $20^{\circ}\text{C}$ . If it takes 5 minutes to reach  $60^{\circ}\text{C}$ , find the time taken to reach  $30^{\circ}\text{C}$ .

## Easier Exam Questions

1. (Fort St 2025 Q13c) Liz heats a mug of milk to  $90^{\circ}\text{C}$  in a microwave. She takes it out into a room where the temperature is a constant of  $26^{\circ}\text{C}$ . The milk cools to  $70^{\circ}\text{C}$  in 5 minutes. At time  $t$  minutes, its temperature  $T$ , in degrees Celsius, decreases according to the equation:

$$\frac{dT}{dt} = -k(T - 26), \text{ where } k \text{ is a positive constant.}$$

- (i) Verify that  $T = 26 + Ae^{-kt}$  is a solution of this equation, where  $A$  is a constant. 1
- (ii) Find the values of  $A$  and  $k$ . 2
- (iii) How long will it take for the temperature to cool to  $30^{\circ}\text{C}$ ? Give your answer correct to the nearest minute. 2
2. (Blacktown Boys 2020 Q12b) Sous vide cooking is a process where food is placed in a vacuum sealed plastic bag or glass jar and immersed in a temperature regulated water bath. A bag of corn, with temperature  $25^{\circ}\text{C}$ , is placed in a water bath that remains at a constant temperature of  $85^{\circ}\text{C}$ . After  $t$  minutes, the temperature of the bag of corn, in degrees Celcius, is  $T$ . The temperature of the water can be modelled using the differential equation

$$\frac{dT}{dt} = k(T - 85)$$

(Do NOT prove this.) where  $k$  is the growth constant.

- (i) After 15 minutes, the temperature of the bag of corn is  $50^{\circ}\text{C}$ . 3  
By solving the differential equation through integration, find the value of  $t$  when the temperature of the bag of corn reaches  $70^{\circ}\text{C}$ . Give your answer to the nearest minute.
- (ii) Sketch the graph of  $T$  as a function of  $t$ . 1
3. (Fort Street 2022 Q13b) The rate of change in temperature,  $T$  degrees celsius, of a metal over time,  $t$  minutes, as it cools is given by the equation  $\frac{dT}{dt} = -k(T - 25)$ . The metal is initially at  $300^{\circ}\text{C}$  and cools to  $250^{\circ}\text{C}$  after 5 minutes.
- i) Show that  $T = 25 + Ae^{-kt}$  is a solution of the equation  $\frac{dT}{dt} = -k(T - 25)$  1
- ii) Find the values of  $A$  and  $k$ . 1
- iii) After how many minutes will the temperature of the metal be  $100^{\circ}\text{C}$ ? Give your answer to the nearest minute. 2
4. (Sydney Grammar 2023 Q12b) The current world population is approximately 8 billion people. Assuming the ‘carrying capacity’ of the earth is 25 billion people, Anna models the population  $P$  after  $t$  years using the differential equation:

$$\frac{dP}{dt} = k(25 - P)$$

- i) Show that  $P = 25 - 17e^{-kt}$  satisfies the differential equation above. 1
- ii) Find  $k$  if the world population after 1 year is 8.072 billion people. Write your answer correct to 3 significant figures. 2
- iii) After how many years does Anna's model say the world population will reach 20 billion people? Round your answer to the nearest whole year. 1
5. (Girraween 2025 Q12b) A freshly baked cake is cooling in a room of constant temperature of  $20^\circ\text{C}$ . At time  $t$  minutes, its temperature  $T$  decreases according to the equation  $\frac{dT}{dt} = -k(T - 20)$  where  $k$  is a positive constant. The initial temperature of the cake is  $150^\circ\text{C}$  and it cools to  $100^\circ\text{C}$  after 15 minutes.
- i) Show that  $T = 20 + Ae^{-kt}$  where  $A$  is a constant is a solution to this equation. 1
- ii) Find the values  $A$  and  $k$ , giving  $k$  correct to 3 decimal places. 2
- iii) How long will it take for the cake to cool to  $25^\circ\text{C}$ ? (Use the value of  $k$  obtained in the previous part) 2
6. (Sydney Boys 2020 Q13a) A model for the population,  $K$ , of kangaroos is

$$K = \frac{21\,000}{7 + 3e^{-\frac{t}{3}}},$$

where  $t$  is the time in years from today.

- (i) Show that  $K$  satisfies the differential equation 2
- $$\frac{dK}{dt} = \frac{1}{3} \left( 1 - \frac{K}{3000} \right) K.$$
- (ii) What is the population today? 1
- (iii) What does the model predict that the eventual population will be? 1
- (iv) What is the annual percentage rate of growth today? 1
7. (Riverview 2021 Q3f) The spread of flu through a student population is modelled by the equation

$$S = \frac{2000}{1 + 199e^{-0.4t}}$$

where  $S$  is the total number of students infected after  $t$  days. Show that the given equation for  $S$  satisfies the differential equation 2

$$\frac{dS}{dt} = \frac{S}{5} \left( 2 - \frac{S}{1000} \right)$$

## Harder Exam Questions

1. (Ascham 2023 Q13d) A dam can sustainably carry 10 million fish.
- The fish population  $N$  (in millions) in the dam follows the logistic differential equation  $\frac{dN}{dt} = \frac{N(10-N)}{200}$  where  $t$  is the time in months since the initial fish count.
- A fish farming company was unaware of this carrying capacity and introduces 15 million fish. Prior to this, the dam did not have fish.
- Find the number of fish remaining after 6 months. Answer to the nearest million fish. 5
- (You may use the identity  $\frac{10}{N(10-N)} = \frac{1}{N} + \frac{1}{10-N}$ .)

2. (Barker 2025 Q11e) A carton of milk is removed from a fridge and placed on a table. The temperature in the fridge is  $3^{\circ}\text{C}$ , and the temperature in the room is  $24^{\circ}\text{C}$ . After sitting on the table for five minutes, the temperature of the milk is  $15^{\circ}\text{C}$ . The temperature ( $T^{\circ}\text{C}$ ) of the milk is given by  $T = B + Ae^{-kt}$ , where  $k$  is a positive constant,  $t$  is time in minutes and  $A$  and  $B$  are constants. Find the total number of minutes that the milk can be left on the table before it reaches a temperature of  $20^{\circ}\text{C}$ . Give your answer correct to the nearest whole number. **3**
3. (Gosford 2022 Q14a) Megan has taken a frozen quiche from the freezer with an initial temperature  $T = 0^{\circ}\text{C}$  and placed it in an oven set to  $200^{\circ}\text{C}$ . After ten minutes she checks the quiche and now it has a temperature of  $T = 20^{\circ}\text{C}$ . The rate that the temperature of the quiche increases is proportional to the difference in temperature between the oven and the quiche.
- (i) Write a differential equation to model this scenario, where  $T$  is the temperature of the quiche and  $t$  is the time in minutes, and use it to show that  $T = 200 + Ae^{-kt}$ . **2**
- (ii) Find the value of  $k$  in exact form. **2**
- (iii) The quiche is ready when the internal temperature reaches  $60^{\circ}\text{C}$ . How much longer after Megan checked the quiche will it be ready? Answer to the nearest minute. **2**
4. (Ascham 2022 Q14a) After 100 snakes were released on an island on 1<sup>st</sup> January 2009 to control the population  $P_1$  of 4000 cane toads, the number of cane toads started to decrease at a rate  $\frac{dP_1}{dt} = -k(P_1 - 20)$  while the population of snakes,  $P_2$ , was changing according to  $\frac{dP_2}{dt} = 0.0005P_2(500 - P_2)$ .
- (i) Given that  $P_1 = 20 + 3980e^{-kt}$  satisfies the differential equation  $\frac{dP_1}{dt} = -k(P_1 - 20)$  and that there was 3119 cane toads left on the island after one year, show that  $k = 0.25$  correct to two decimal places. **1**
- (ii) Given that  $\frac{1}{0.0005P_2(500 - P_2)} = 4\left(\frac{1}{P_2} + \frac{1}{500 - P_2}\right)$  show that the equation of the population  $P_2$  of snakes is given by  $P_2 = \frac{500}{1 + 4e^{-0.25t}}$ . **3**
- (iii) Assuming that  $P_1 = 20 + 3980e^{-0.25t}$ , in what year will the population of snakes exceed the population of cane toads? **2**
5. (Sydney Girls 2022 Q12a) The rate at which a body warms in air is proportional to the difference between its temperature  $T$  and the constant temperature  $A$  of the surrounding air. This rate can be expressed using the differential equation:

$$\frac{dT}{dt} = k(T - A)$$

where  $t$  is the time in minutes and  $k$  is a constant.

- (i) Show that  $T = A + Pe^{kt}$ , where  $P$  is a constant, is a solution of the differential equation. **1**
- (ii) At the time a cheesecake is removed from a fridge, it is  $2^{\circ}\text{C}$ . **3**  
 After 15 minutes, it has warmed to  $10^{\circ}\text{C}$ .  
 Given the air temperature around the cheesecake is  $25^{\circ}\text{C}$ , find the temperature of the cheesecake after a further 15 minutes.  
 Give your answer in degrees, correct to two decimal places.
6. (Gosford 2020 Q14c) A tank contains 2500 litres of water and 25 Kg of dissolved salt. Fresh water enters the tank at a rate of 20 litres per minute. The solution is thoroughly mixed at all times and is drained from the tank at a rate of 15 litres per minute.

(i) Using  $y$  for the amount of salt in the tank in kilograms (as a function of time), and  $t$  for time in minutes, show that the concentrations of salt in the tank at time  $t$  can be given by  $C = \frac{y}{2500+5t}$  1

(ii) Explain why the rate of change of salt in the tank can be given by  $y' = -15C$  1

(iii) Find  $y$ , the amount of salt in the tank as a function of  $t$ . 4

7. (Girraween 2022 Q15a) A heated metal ball is dropped into a liquid.  
As the ball cools, its temperature,  $T^\circ \text{C}$ ,  $t$  minutes after it enters the liquid is given by:

$$T = 25 + 400e^{-0.05t}, t \geq 0$$

(i) Find the temperature of the ball as it enters the liquid. 1

(ii) Find the value of  $t$  when  $T = 300$ .  
Answer correct to 3 significant figures. 1

(iii) Find the rate at which the temperature of the ball is decreasing at the instant when  $t = 50$ .  
Give your answer in  $^\circ\text{C}$  per minute to 3 significant figures. 2

(iv) Using the equation given above, explain why the temperature of the ball can never fall to  $20^\circ \text{C}$  1

8. (Ascham 2024 Q13d) The differential equation for the repopulation of koalas in a particular habitat is given by

$$\frac{dP}{dt} = \frac{P(300 - P)}{300}$$

where  $P$  is the population of koalas and  $t$  is the number of years. Initially 120 koalas are released.

(i) Using the identity  $\frac{300}{P(300-P)} = \frac{1}{P} + \frac{1}{300-P}$  (without proof) show that 3

$$P = \frac{600}{2 + 3e^{-t}}.$$

(ii) This population of koalas reaches its maximum growth rate within the first year. 2  
After how many complete months will the growth rate start decreasing? Justify your answer.

9. (Riverview 2020 Q14b) The area  $A \text{ cm}^2$  is occupied by a bacterial colony. The colony has its growth modelled by the logistic equation  $\frac{dA}{dt} = \frac{1}{25}A(50 - A)$  where  $t \geq 0$  and  $t$  is measured in days. At time  $t = 0$ , the area occupied by the bacteria colony is  $\frac{1}{2} \text{ cm}^2$ .

(i) Show that  $\frac{1}{A(50-A)} = \frac{1}{50} \left( \frac{1}{A} + \frac{1}{50-A} \right)$ . 2

(ii) Using the result from part (b) (i), solve the logistic equation and hence show that  $A = \frac{50}{1+99e^{-2t}}$ . 3

(iii) According to this model, what is the limiting area of the bacteria colony? 2

10. (Caringbah 2021 Q13c) A freshly caught fish, initially at  $18^\circ \text{C}$ , is placed in a freezer that has a constant unknown temperature of  $x^\circ \text{C}$ . The cooling rate of the fish is proportional to the difference between the temperature of the freezer and the temperature  $T^\circ \text{C}$ , of the fish.

It is known that  $T$  satisfies the equation  $\frac{dT}{dt} = -k(T - x)$ ,

where  $t$  is the number of minutes after the fish has been placed in the freezer.

- (i) Show that  $T = x + Ae^{-kt}$  satisfies this equation where  $A$  is a constant. **1**
- (ii) If the temperature of the fish is  $10^\circ\text{C}$  after  $7\frac{1}{2}$  minutes, show that the fish's temperature after  $t$  minutes is given by  $T = x + (18 - x)e^{\frac{2}{15} \log_e \left[ \frac{10-x}{18-x} \right] t}$  **3**
- (iii) Find the temperature of the fish after 15 minutes when the freezer temperature is  $5^\circ\text{C}$ . Answer to the nearest degree. **1**

11. (Penrith 2025 Q11c) Eleanor placed an object in a freezer to cool. After  $t$  minutes, its temperature is  $T^\circ\text{C}$ . The freezer is at a constant temperature of  $-8^\circ\text{C}$ .

The object's temperature decreases according to the differential equation  $\frac{dT}{dt} = k(T + 8)$ .

- (i) If the object's initial temperature is  $40^\circ\text{C}$ , solve the differential equation for  $T$  in terms of  $k$  and  $t$ . **2**
- (ii) If the object cools to  $30^\circ\text{C}$  in half an hour, show that  $k = \frac{1}{30} \ln \left( \frac{19}{24} \right)$ . **1**
- (iii) When will the temperature of the object be  $0^\circ\text{C}$  correct to the nearest minute? **2**